Sanskar Rajesh Mankar

Quantitative Researcher | Systematic Strategies & Options Backtesting
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SUMMARY

Quantitative researcher with 3 years of experience building systematic strategies on NIFTY indices and BTC, and the tooling to test whether an edge is real. Developed Hawkes-process intraday momentum strategies (Sharpe 2.91 on NIFTY 50, 3.83 on NIFTY Next 50) and **Option Lab**, an open-source options backtesting engine that prices every leg from 1-minute chain data with real exchange costs, slippage and Monte Carlo drawdowns. Validation-first: no look-ahead, out-of-sample and walk-forward splits, parameter-stability checks. Prior end-to-end ML engineering (LLM fine-tuning, RAG, computer vision).

SELECTED RESEARCH

Option Lab: Options Backtesting Engine | Python, Streamlit | Case study | Open source

- Turns any signal into multi-leg option strategies and prices each leg from **1-minute NIFTY option-chain data (1,255 trading days)** instead of using the index as a proxy.
- Charges STT, exchange, SEBI, stamp duty, GST and slippage on every fill; supports stops, linked exits and Monte Carlo reshuffles of daily P&L to size drawdown risk.
- Parallel optimiser with result caching: **~125x faster**, **80k parameter combinations per run**; 4.6k lines of Python. Case-study strategy: 33.3% CAGR.

Hawkes Intraday Momentum (Long/Short) | NIFTY 50, NIFTY Next 50 | Python

- A Hawkes-intensity filter keeps the strategy out of sideways sessions and in the trend when volatility bursts. Built on NIFTY 50; the same logic held up on higher-beta Next 50.
- **NIFTY Next 50:** 46.6% CAGR, Sharpe 3.83, max drawdown -4.2%, 2,325 trades (2019-2025). **NIFTY 50:** 31.7% CAGR, Sharpe 2.91, max drawdown -5.9%.

Overnight Drift (BTST), Long-Only | NIFTY 50, Midcap Select, Next 50 | Python, Pine Script

- **NIFTY 50:** 946 trades, 59.8% win rate, 22.95% CAGR, Sharpe 3.21, Sortino 6.20, profit factor 1.90, max drawdown -3.81%. Approx. 25% CAGR on Midcap Select, 30%+ on Next 50.
- Screened 300+ equities and showed the edge does **not** survive cash-equity costs, so scoped it to index instruments.

Hawkes Volatility Breakout on BTC | Python | Ran live on Untrade

- Detects volatility expansion from liquidation-driven clustering and predicts direction. Backtest: 75% CAGR, Sharpe 3.2, max drawdown -16%; ran live and ranked among the platform's top performers.

PROFESSIONAL EXPERIENCE

Independent Quantitative Researcher | Self-Employed, Remote — 07/2025 – Present

- Research and backtest systematic strategies on Indian indices and equities, testing each edge across instruments and market regimes.
- Built in-house infrastructure: event-driven backtesting, look-ahead prevention, out-of-sample and walk-forward testing, Monte Carlo robustness and parameter-stability analysis.
- Created Option Lab after an early unhedged options loss showed that backtesting the index is not backtesting the traded instrument; now every strategy is tested on the exact instrument with full costs.
- Current research: stock swing-trading systems and BTC options strategies.

Quantitative Researcher | Untrade, Jaipur, India — 02/2024 – 07/2025

- Developed a live Hawkes-process BTC alpha (volatility expansion and direction); built Ethereum alpha models using event-driven signals and systematic parameter optimisation.
- Engineered a 5-island genetic algorithm with adaptive mutation and parallel evaluation (ThreadPoolExecutor) for faster parameter exploration.
- Wrote research protocols for look-ahead detection, strict in-sample/out-of-sample validation and Monte Carlo testing; automated the pipeline from data extraction to risk analysis.

AI/ML Engineer Intern | Rajyug Solutions, Pune, India — 09/2023 – 01/2024

- Fine-tuned an 8B-parameter model for medical Q&A with CUDA-accelerated inference; deployed ML applications as FastAPI services.
- Built a FAISS-based RAG pipeline grounded in source PDFs; developed a YOLOv8 helmet detector (97% accuracy), plus emotion-classification and resume-screening models.

TECHNICAL SKILLS

Quant Research	Alpha research, Hawkes processes, volatility clustering, time-series analysis, options pricing & multi-leg strategies, genetic algorithms, Monte Carlo simulation
Validation	Event-driven backtesting, out-of-sample and walk-forward testing, look-ahead bias prevention, parameter-stability analysis, transaction-cost modelling, risk analysis
Programming	Python, Pine Script, NumPy, Pandas, Streamlit, parallel processing (ThreadPoolExecutor), OHLCV and 1-minute option-chain pipelines
Machine Learning	PyTorch, LLM fine-tuning, RAG (FAISS), YOLOv8, CUDA, FastAPI

EDUCATION

B.Tech, Information Technology | MIT ADT University, Pune — 2023

All performance figures are historical backtests, not live trading, unless stated otherwise.